Identifying Breast Cancer Screening Gaps and Mortality Disparities in Rural Michigan Counties

Abstract

This analysis examines how population health data can be used to identify Michigan counties with disparities in breast cancer mortality. Using county-level mortality data alongside socioeconomic factors and regional screening rates, patterns were explored through exploratory data analysis and correlation mapping. The results show that mortality is influenced by multiple factors, including age and socioeconomic conditions, and is not explained by screening alone.

To better identify areas for intervention, a combined measure of mortality rate and total deaths was used to prioritize counties based on both severity of outcomes and population-level impact. This approach highlights two distinct patterns of need: high-burden urban counties and high-rate rural counties. These findings suggest that improving outcomes requires focusing not only on access to screening, but also on follow-up care, care coordination, and access to treatment services.

1. Introduction

Breast cancer remains a significant public health concern, and early detection through routine mammography screening is one of the most effective ways to improve outcomes. However, access to preventive care and resulting health outcomes can vary widely, particularly in rural areas. This analysis focuses on Michigan counties to better understand where gaps in screening and higher mortality rates exist, and how social and demographic factors contribute to those differences. The goal is to move beyond simple comparisons and use population health analytics to identify where targeted outreach and intervention efforts would be most effective.

2. Data and Methodology

Data Sources

Category	Source	Metrics Used
Mortality	CDC WONDER (ICD-10 C50)	County-level crude rates
Screening	Michigan Cancer Atlas	Prosperity Region mammography rates
Socioeconomics	Michigan Cancer Atlas	Poverty, Uninsured, Education, Food Insecurity

Data

The primary outcome variable is breast cancer mortality (crude rate). The primary exposure is the percent of women receiving mammograms. Additional explanatory variables include poverty rate, uninsured rate, education level, food insecurity, unemployment, and percent of residents aged 65 and older. Age was included as a contextual variable rather than a filter.

The data used in this analysis includes breast cancer mortality data from the Michigan Cancer Atlas (2016–2022) and CDC WONDER (2018–2022), along with socioeconomic and demographic variables derived from the Michigan Cancer Atlas. Additionally, the data reflects aggregated multi-year estimates across slightly different reporting periods, which may smooth short-term variation but provides a more stable view of underlying patterns.

Data Preparation

Datasets were merged using county and Prosperity Region mappings to align mortality, screening, and socioeconomic variables. Numeric variables were cleaned and standardized to ensure consistency. Some counties were missing due to data suppression related to small population sizes and were excluded from analysis.

Analytical Approach

1. **Exploratory Data Analysis (EDA):** Used summary statistics and histograms to identify outliers and age-distribution skewness.
2. **Correlation Mapping:** Examined the strength of relationships between mortality, screening, and socioeconomic variables.
3. **Quadrant Analysis:** Counties were categorized based on median-based thresholds across two dimensions:
    - *Screening vs. Mortality*
    - *Poverty vs. Screening*
4. **Overlap Analysis:** Identified "high-risk" counties appearing in multiple negative quadrants. While this provided initial directional insight, it was not used as the primary method for identifying priority counties due to limitations in screening data.

Note on Limitations: Screening data was only available at the regional level, which limits the ability to detect true county-level variation in screening rates. Mortality rates are based on crude values and may be influenced by differences in age distribution across counties. Additionally, smaller counties may show greater variability due to population size.

3. Key Findings

Demographic & Socioeconomic Drivers

- **The Age Factor:** Mortality rates vary across counties, with most clustering within a moderate range and a small number of counties appearing as outliers with higher mortality. Age distribution is right-skewed, indicating that some counties have substantially older populations, which is important when interpreting mortality differences.

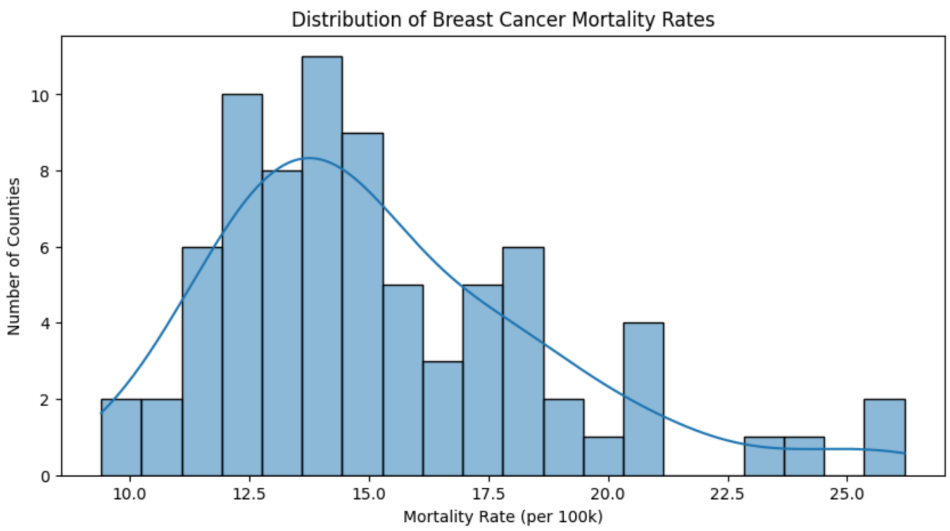

Figure 1. Distribution of Breast Cancer Mortality Rates

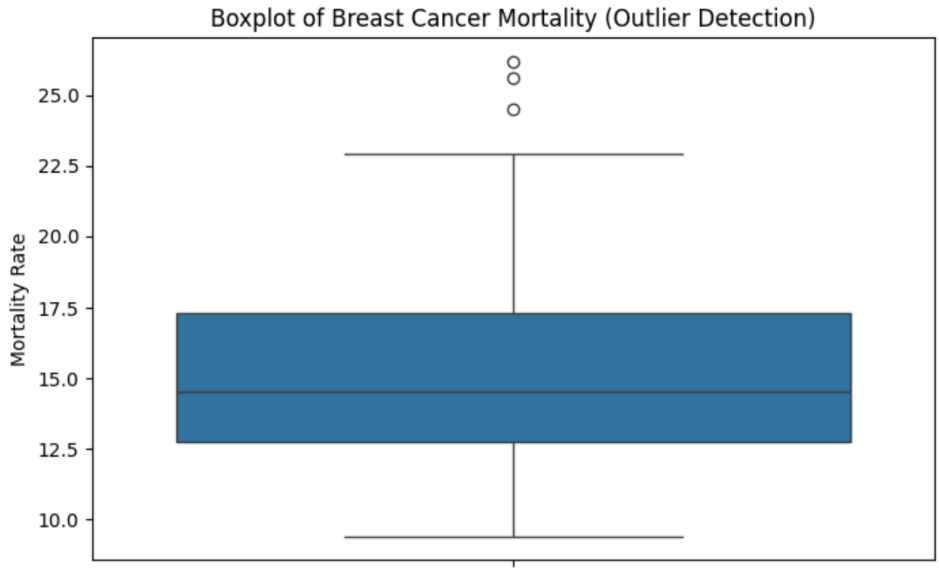

Figure 2. Breast Cancer Mortality Boxplot (Outliers & Spread)

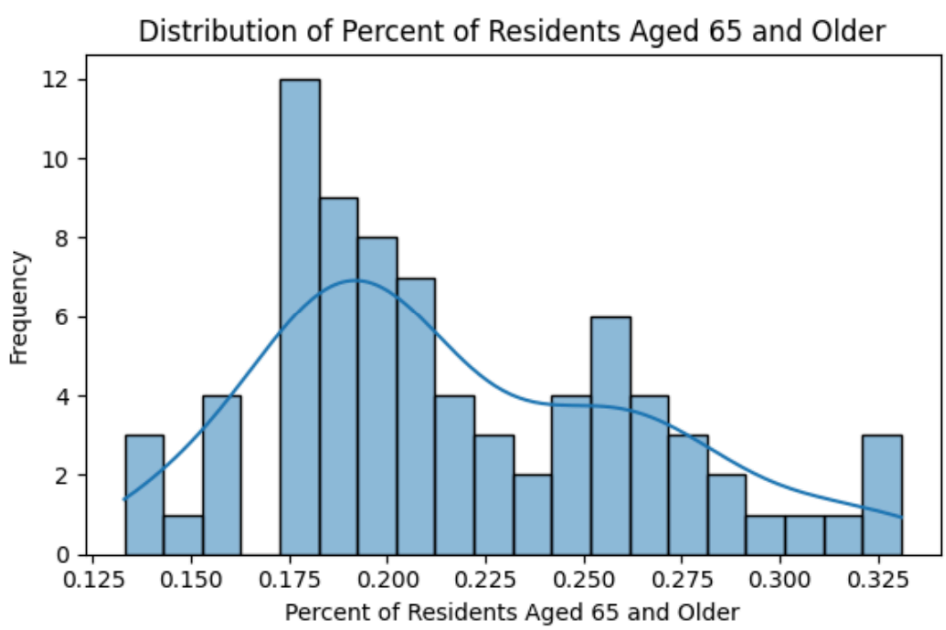

Figure 3. Distribution of Population Aged 65 and Older

- **Socioeconomic Clustering:** Poverty, food insecurity, and unemployment are highly correlated, suggesting that health challenges in Michigan tend to cluster in specific "high-burden" pockets.

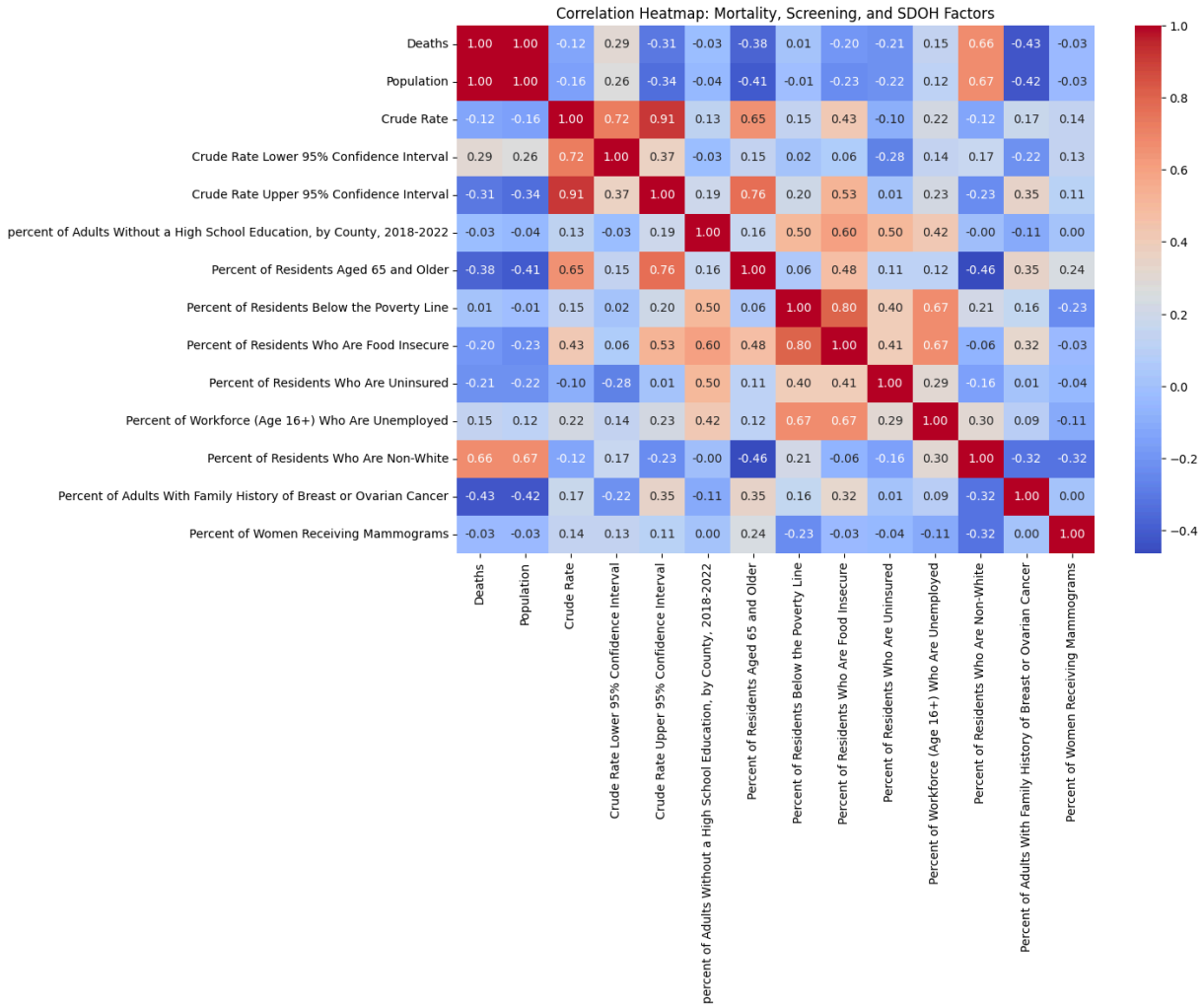

Figure 4. Correlation Between Mortality, Screening, and SDOH Factors

	High Mortality Mean	Low Mortality Mean	Difference
age_65_plus	0.244	0.191	0.053
food_insecure	0.132	0.116	0.016
screening	0.800	0.792	0.008
poverty	0.136	0.128	0.008
unemployed	0.061	0.054	0.007
family_history	0.364	0.357	0.007
no_high_school	0.092	0.087	0.005
uninsured	0.059	0.061	-0.002
nonwhite	0.134	0.145	-0.011

Table 1. Comparison of High and Low Mortality Counties

A comparison between high and low mortality counties shows that age has the largest difference between groups, followed by food insecurity and poverty. This reinforces that mortality disparities are more strongly associated with demographic and socioeconomic factors than with screening rates alone.

- **Screening Disconnect:** Consistent with the comparison above, high screening rates did not always correlate with lower mortality at the county level. In contrast, age shows a stronger association with mortality, indicating that demographic structure plays a significant role. Socioeconomic variables such as poverty, food insecurity, and unemployment are strongly correlated with each other, suggesting that these challenges tend to cluster within the same counties.

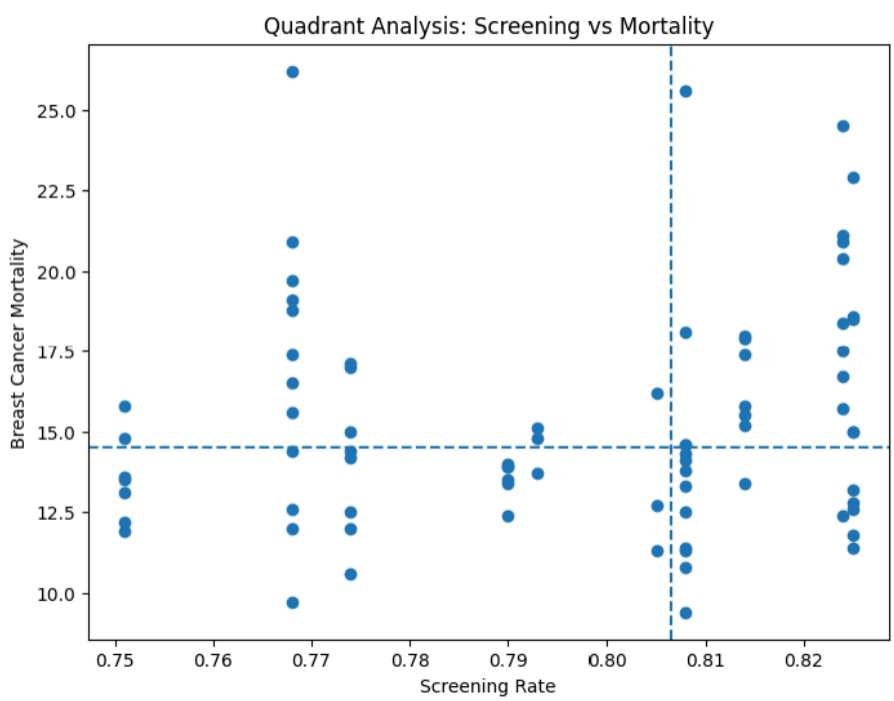

Figure 5. Screening vs Mortality Quadrant Analysis

Note: Screening rates are measured at the regional level; mortality is measured at the county level

	screening
poverty	-0.234854
unemployed	-0.111152
uninsured	-0.040930
food_insecure	-0.029231
no_high_school	0.003796
age_65_plus	0.241640
screening	1.000000

Table 2. Correlations Between Screening and Access-Related Variables

Correlation analysis shows a modest negative relationship between screening and poverty and a modest positive relationship between screening and age. In your saved output, screening has a negative correlation with poverty (-0.235) and unemployment (-0.111), and a positive correlation with age 65+ (+0.242), while the relationship with uninsured and food insecurity is weak.

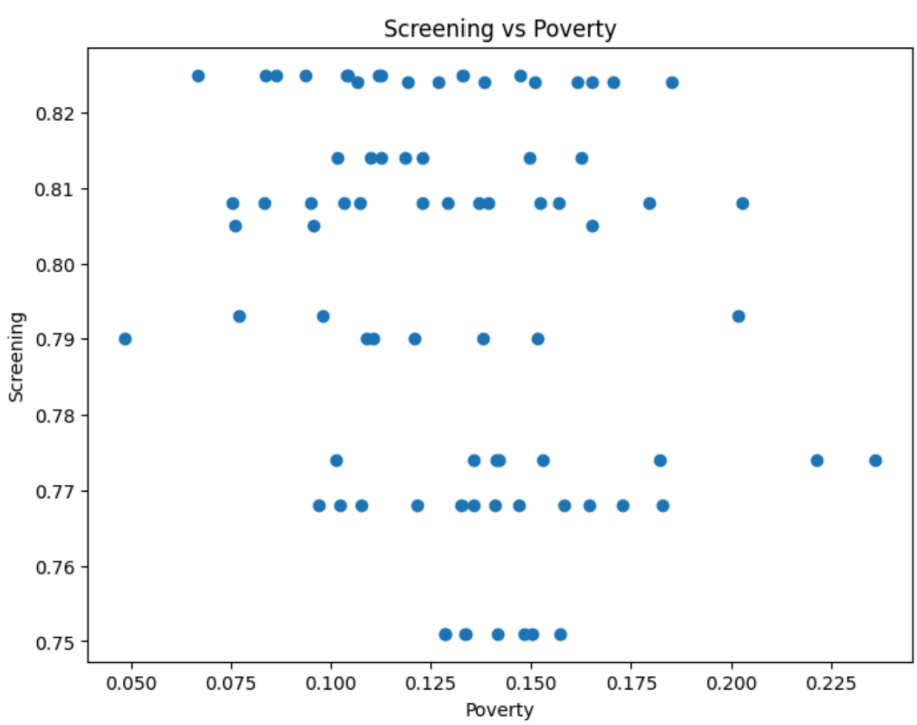

Figure 6. Poverty vs Screening Relationship

Regional Patterns

There are clear differences across Prosperity Regions in both mortality and poverty. Some regions show consistently higher mortality or greater variability across counties, while others are more uniform.

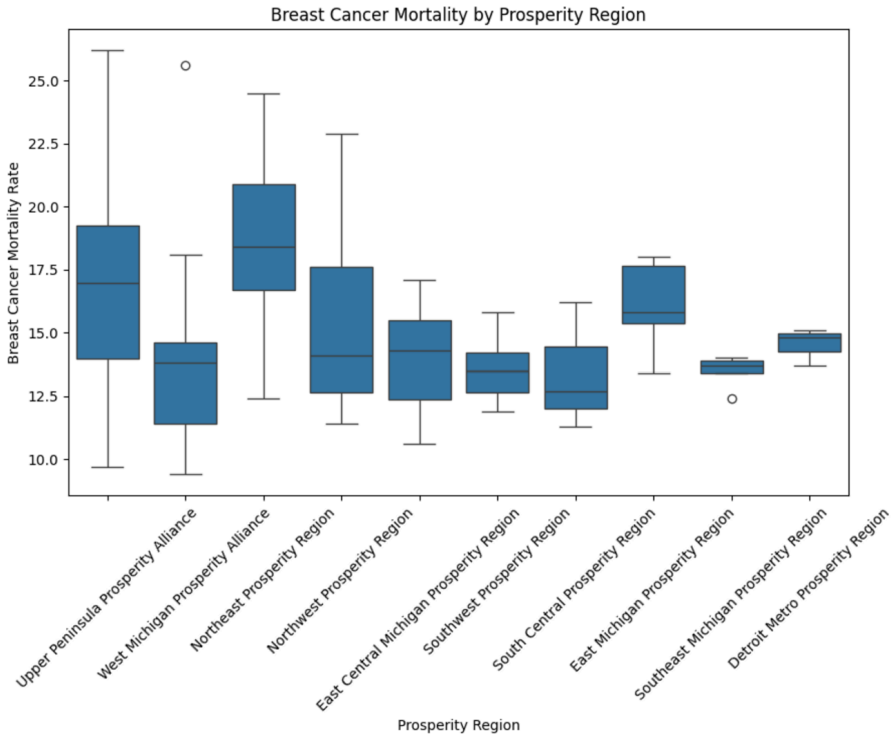

Figure 7. Mortality by Prosperity Region

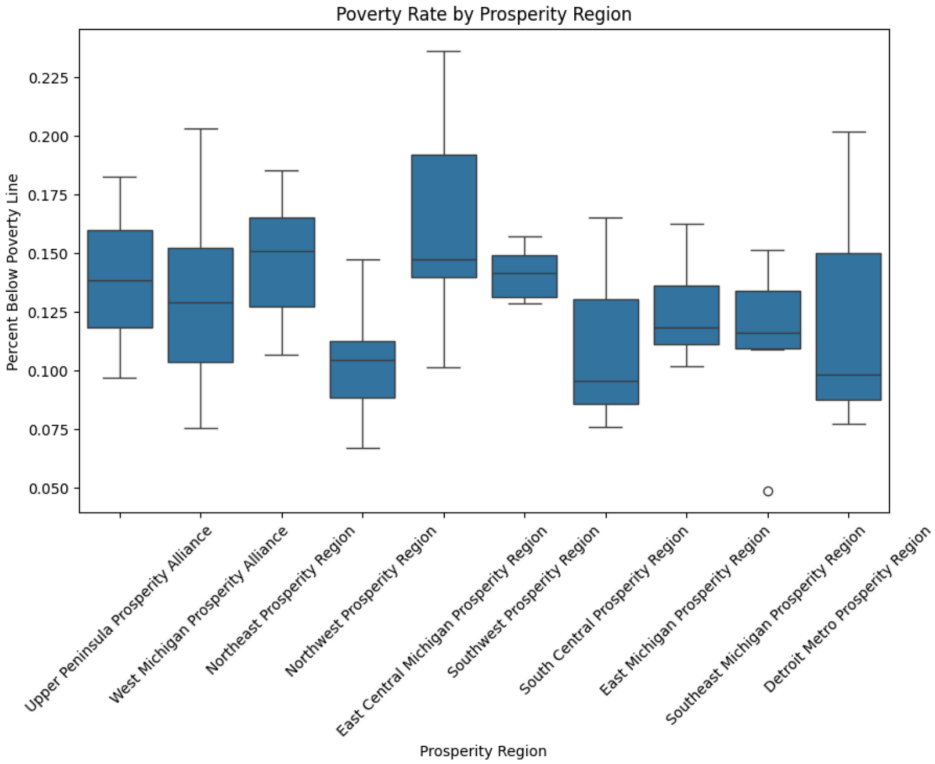

Figure 8. Poverty by Prosperity Region

Prosperity Region	mortality	screening	poverty	uninsured	food_insecure	unemployed	age_65_plus
Northeast Prosperity Region	18.622	0.824	0.147	0.060	0.147	0.065	0.282
Upper Peninsula Prosperity Alliance	16.908	0.768	0.139	0.055	0.132	0.060	0.242
East Michigan Prosperity Region	16.171	0.814	0.125	0.056	0.124	0.061	0.207
Northwest Prosperity Region	15.180	0.825	0.104	0.063	0.114	0.048	0.245
Detroit Metro Prosperity Region	14.533	0.793	0.126	0.050	0.110	0.065	0.169
West Michigan Prosperity Alliance	14.138	0.808	0.130	0.066	0.117	0.053	0.193
East Central Michigan Prosperity Region	14.100	0.774	0.164	0.066	0.139	0.065	0.209
Southwest Prosperity Region	13.557	0.751	0.141	0.071	0.118	0.057	0.188
Southeast Michigan Prosperity Region	13.517	0.790	0.113	0.048	0.106	0.051	0.184
South Central Prosperity Region	13.400	0.805	0.112	0.045	0.104	0.050	0.171

Table 3. Average Mortality, Screening, and SDOH Factors by Prosperity Region

Regional averages provide context behind the boxplots and show that mortality, screening, poverty, and age differ meaningfully across regions. In your saved output, the Northeast Prosperity Region has the highest average mortality (18.622), while the Southwest and Southeast regions are among the lowest. East Central shows the highest average poverty (0.164), and the Northwest and Northeast regions show some of the highest age 65+ values .

- **Detroit Metro:** High internal disparity; extreme variations in poverty despite being a single prosperity region.
- **Rural Variability:** Rural counties showed the highest fluctuations in mortality, likely due to a combination of aging populations and limited access to specialized care.

4. Priority Counties for Intervention

The initial quadrant analysis focused on identifying counties with **low screening and high mortality** as potential gaps in preventive care.

However, screening data was only available at the regional level and showed limited variation across counties. As a result, screening alone did not reliably explain differences in mortality outcomes.

To better identify areas for intervention, the approach was refined to focus on two key dimensions:

- **Mortality Rate (Crude Rate):** Indicates severity of outcomes
- **Total Deaths (Burden):** Reflects population-level impact

This combined approach allows for identification of counties that are not only experiencing worse outcomes, but also contributing the greatest overall impact.

Updated Priority County Identification Method

Counties were evaluated using a combined priority score based on mortality rates and total deaths, which were standardized using z-scores and equally weighted to capture both severity of outcomes and overall population impact.

This approach identifies two types of high-need areas:

- High-burden urban counties, where large populations drive total deaths
- High-rate rural counties, where smaller populations experience disproportionately high mortality

Priority Counties

The following counties were identified as highest priority based on combined mortality rate and burden:

- **Wayne** – Highest total deaths with elevated mortality
- **Oakland** – High population burden with moderate mortality
- **Macomb** – High burden with above-average mortality
- **Luce** – Extremely high mortality rate despite small population
- **Lake** – High mortality rate with low population size

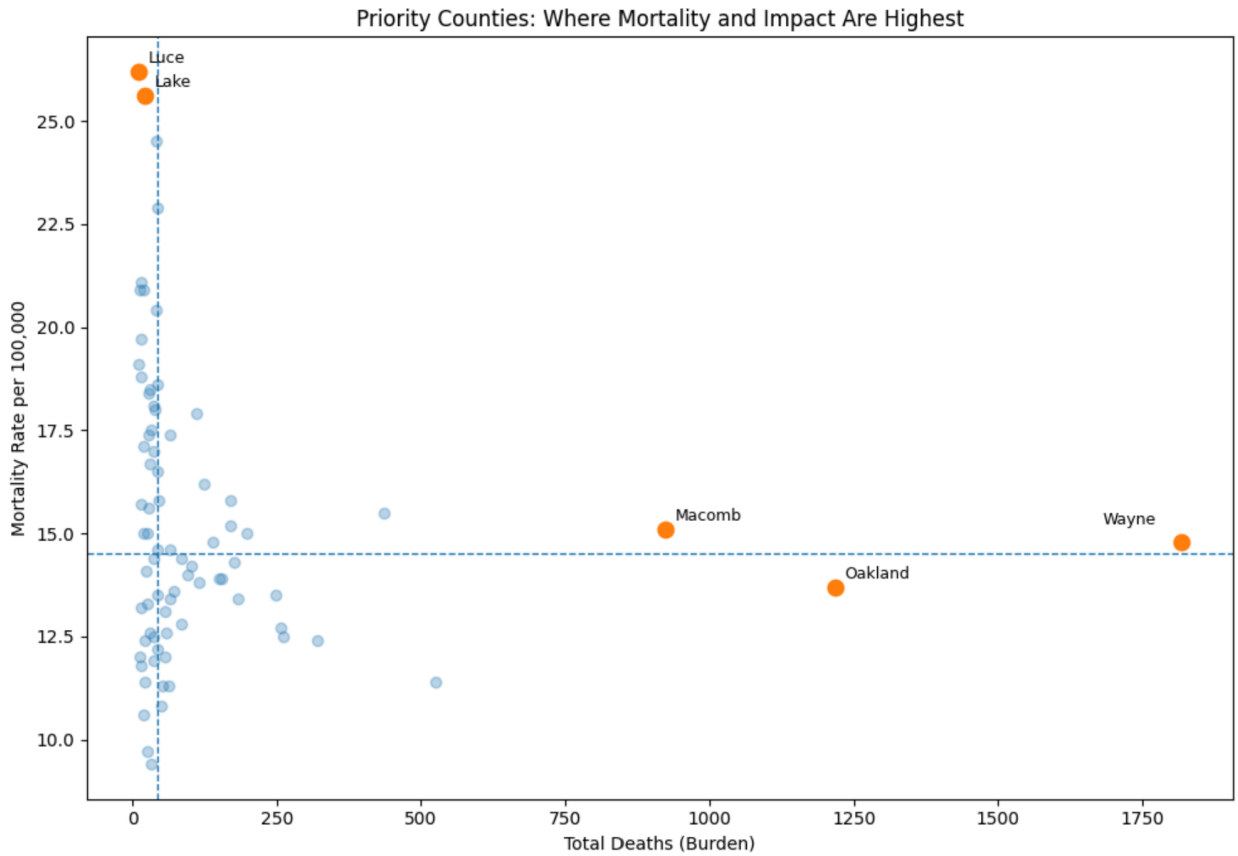

Figure 9. Priority Counties Based on Mortality Rate and Burden

county	population	deaths	mortality_rate	priority_score
Wayne	12,297,950	1,818	14.8	3.085
Oakland	8,877,006	1,219	13.7	1.811
Macomb	6,131,785	924	15.1	1.461
Luce	42,040	11	26.2	1.341
Lake	85,962	22	25.6	1.276
Roscommon	167,025	41	24.5	1.155
Charlevoix	183,135	42	22.9	0.929
Montmorency	66,302	14	21.1	0.620
Genesee	2,827,491	437	15.5	0.610
Presque Isle	90,964	19	20.9	0.600

Table 4. Priority Counties for Intervention

**Priority score reflects both severity (mortality rate) and impact (total deaths), standardized and equally weighted.*

5. Conclusion & Recommendations

Key Takeaways

Breast cancer mortality varies across Michigan counties and is not explained by screening rates alone. While screening remains important, outcomes are more strongly associated with demographic and socioeconomic factors, as well as differences in access to follow-up and treatment.

Using a combined approach that incorporates both mortality rate and total deaths, this analysis identifies two distinct patterns of need. Urban counties such as Wayne, Oakland, and Macomb contribute the greatest overall burden due to population size, while smaller rural counties such as Luce and Lake experience disproportionately high mortality rates.

The top five counties account for approximately 40% of total breast cancer deaths, highlighting the importance of addressing high-burden areas alongside high-rate rural counties.

These findings highlight that disparities exist across different contexts and require targeted strategies. Addressing mortality requires moving beyond screening access and focusing on what happens after detection, including care coordination, timeliness of follow-up, and access to treatment services.

Focusing on only one metric would miss part of the problem, as high-burden and high-rate counties represent different types of need.

Strategic Recommendations

The findings from this analysis suggest that breast cancer mortality disparities across Michigan counties are not driven by screening rates alone, but are more strongly associated with demographic and socioeconomic factors. As a result, interventions should focus on improving what happens after initial access to care rather than expanding screening alone.

1. Measure and Improve Follow-Up After Abnormal Screening

Rather than assuming that access to screening is the primary issue, a more actionable step is to evaluate how quickly patients receive follow-up care after an abnormal mammogram. Established quality measures, such as follow-up within 90 days, can be used to assess whether patients are completing diagnostic imaging or biopsy in a timely manner.

Health systems and Medicaid plans operating in Michigan can use this type of measure to identify gaps in care coordination, particularly in counties with higher mortality rates. This approach focuses on improving outcomes using existing data and infrastructure rather than introducing new programs without measurable impact.

2. Strengthen Referral Pathways to Regional Cancer Centers

Advanced cancer care in Michigan is concentrated in a limited number of regional hubs, including Detroit, Ann Arbor, Grand Rapids, and several smaller regional centers. Counties identified in this analysis as high-risk may not lack screening access, but may instead face challenges in accessing diagnostic and specialty care after an abnormal result.

A practical step is to improve referral coordination between local providers and regional cancer centers. This can include standardized referral workflows, faster scheduling for high-risk patients, and clearer care pathways following abnormal screening results. These improvements can reduce delays without requiring large-scale infrastructure changes.

3. Evaluate Geographic Access to Diagnostic and Treatment Services

This analysis highlights that counties with higher mortality often overlap with areas that are more geographically distant from major treatment hubs. While this report does not directly measure travel time or provider availability, the findings suggest that geographic access may play a role in delayed diagnosis or treatment.

As a next step, health systems or state agencies could map travel distance to the nearest cancer treatment facility and compare it to outcomes. This would provide a more concrete understanding of whether geographic barriers are contributing to disparities and help prioritize regions for targeted interventions.

Future analysis should quantify travel time to the nearest cancer treatment facility and evaluate its relationship with mortality outcomes to better understand the role of geographic barriers.

4. Prioritize Targeted Interventions in High-Risk Counties

Resources should be directed toward counties identified through the combined mortality rate and burden analysis. These counties represent areas with either a high number of deaths, high mortality rates, or both.

This includes both high-burden urban counties and high-rate rural counties, ensuring that interventions address both population-level impact and severity of outcomes.

5. Use Existing Data to Monitor Outcomes and Guide Action

This analysis demonstrates the value of combining mortality, screening, and socioeconomic data to identify patterns. Moving forward, similar datasets can be used to monitor trends over time, evaluate the effectiveness of interventions, and refine targeting strategies.

Rather than relying on assumptions about access or behavior, ongoing analysis should focus on measurable indicators such as time to diagnosis, stage at diagnosis, and regional variation in outcomes.

Final Statement

Population health analytics provides a structured way to move from identifying patterns to informing action. By combining mortality, screening, and socioeconomic data, this analysis highlights where interventions can have the greatest impact.

Critically, improving outcomes requires moving beyond access to screening and focusing on what happens after detection, including follow-up care, care coordination, and access to treatment services. These factors are likely key drivers of the disparities observed across both high-burden and high-rate counties.